Scaled High Temperature Test Reactor Cogeneration for Zero-Emission Hybrid Desalination with Water Independent Heat Rejection

Imperial Team 5

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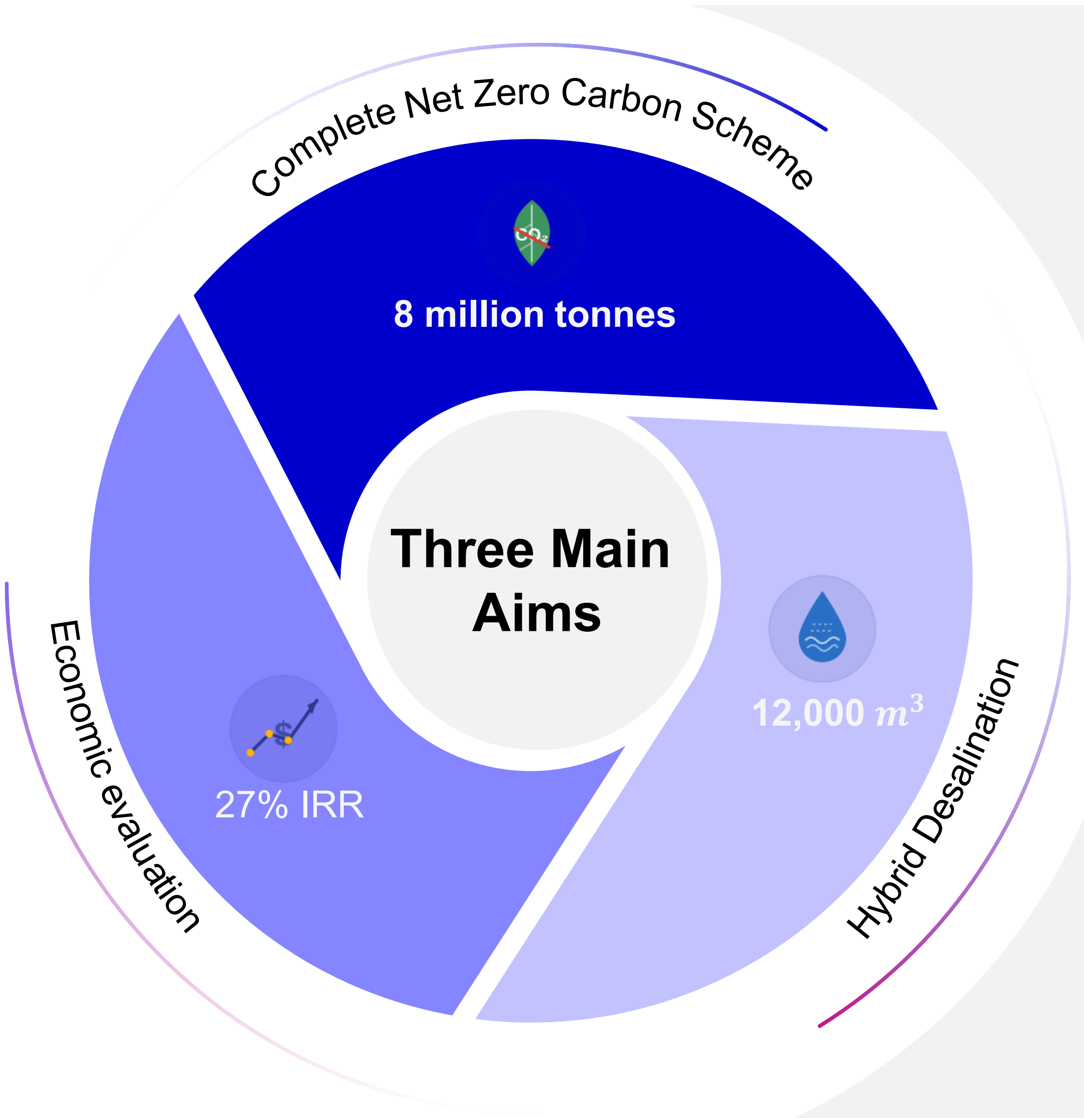
PART I: HTGR Contents

1. Project Scope and Timeline



Net zero carbon scheme

- Enough to produce electricity and power for a large city up to 100 years
- Alternative to fossil-fuel Desalination
- CO_2 offset from electrical and thermal power

20% ▲

Economic Evaluation

- 27% IRR
- 4.3 Year Payback

95% ▲

Secondary Application

- Multi-Effect Desalination with Thermal Vapour Compression
- Reverse Osmosis
- Cold Seawater feed advantage

33% ▲

2. Literature Review
2.1. Down Selection Process and Key Metric Reasoning

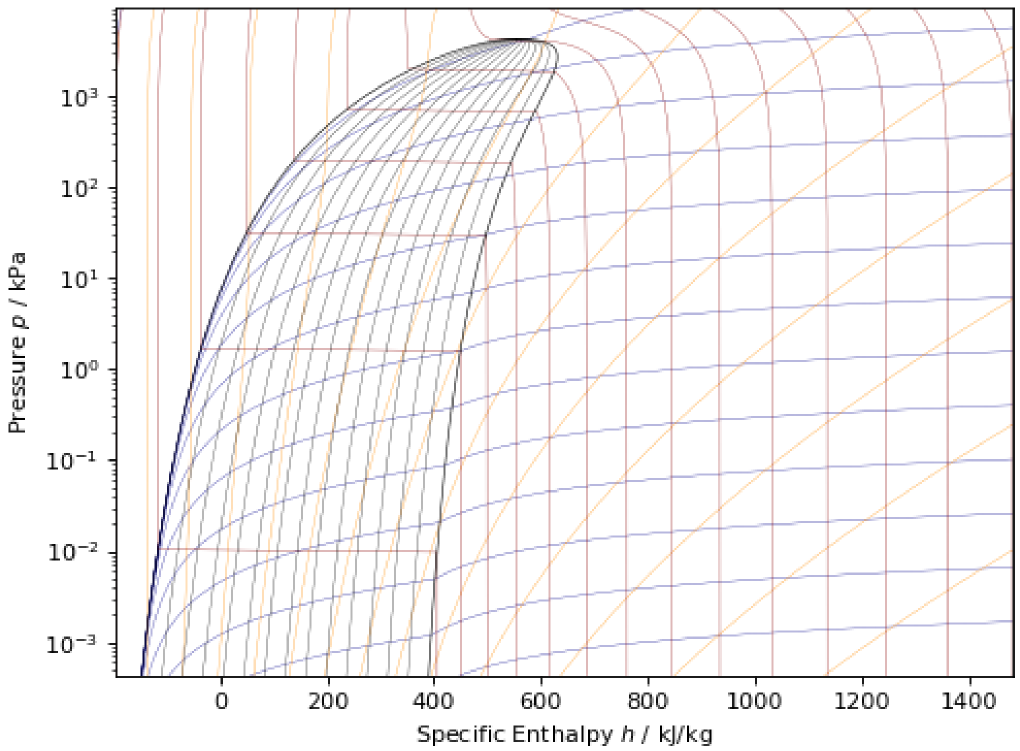
Reactor Type	Key Parameters	Advantages	Limitations
Pebble-Bed HTGR	• 200–300 MWth • 70–80 MWe • 750°C outlet	• Proven reactor architecture • Industrial scale output	• Complex systems • Not optimised for MED desalination
Micro-HTGR	• 0.5–20 MWth • 0.1–5 MWe • 750°C outlet	• Very passive safety • Portable/ remote use	• Very small output • Limited desalination capability
HTTR	• 300 MWth • 30–50 MWe • 950°C	• Ideal for MED Desalination • Air-cooled • High stability and safety	• Research-derived design

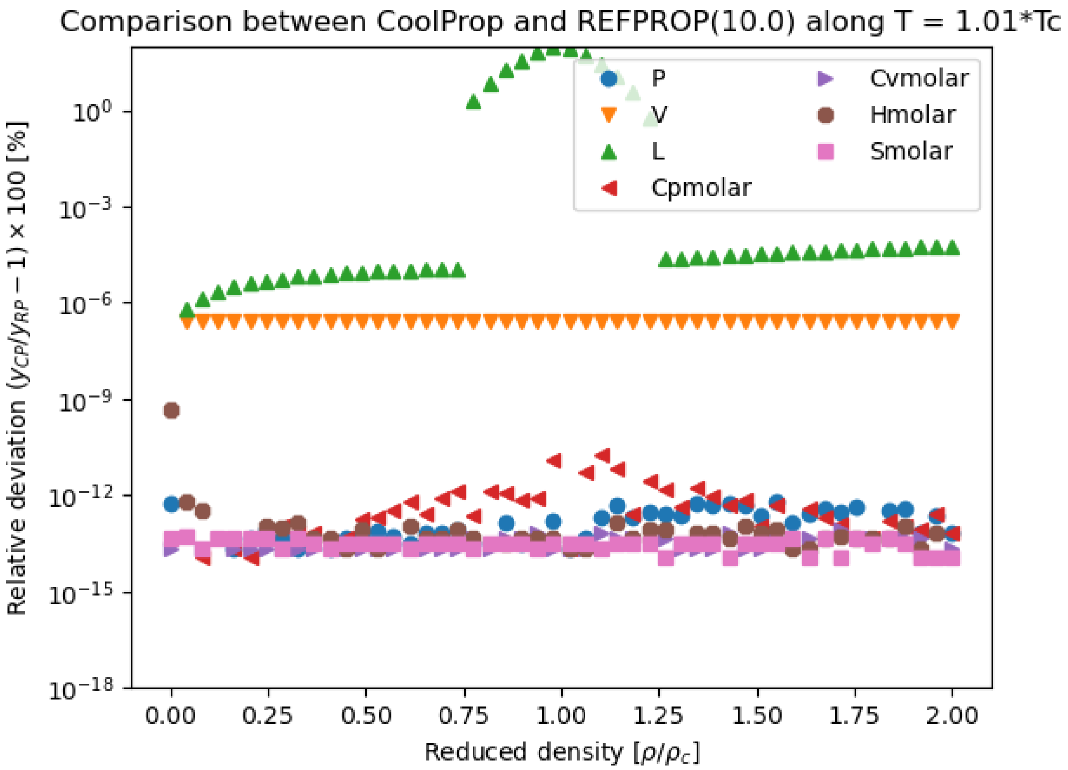
Metric	HTTR-Desalination Values
Reactor Thermal Power	300 MWth
MED GOR	10–12
Water Production	7,341–15,256 m^3/day (theoretical)
Cooling Requirement	Very low
Electricity Output	30–50 MWe

Reason Category	HTTR-Desalination Advantage
High-temperature output	950 °C design → supports high-efficiency MED/TVC
Right-sized thermal power	300 MWth matches a semi-industry scale desalination plant
Water production capability	9–12k m^3/day potable water reliably
Electricity cogeneration	30–50 MWe from small Rankine/H_2O cycle
Cooling method	Liquid metal and air-cooled condensers at high temperature; radiative cooling < 200 °C
Safety	Strong passive characteristics; high heat capacity graphite
Community suitability	Small footprint; minimal marine impact; stable 24/7 output
Independent heat rejection power	148 MWth

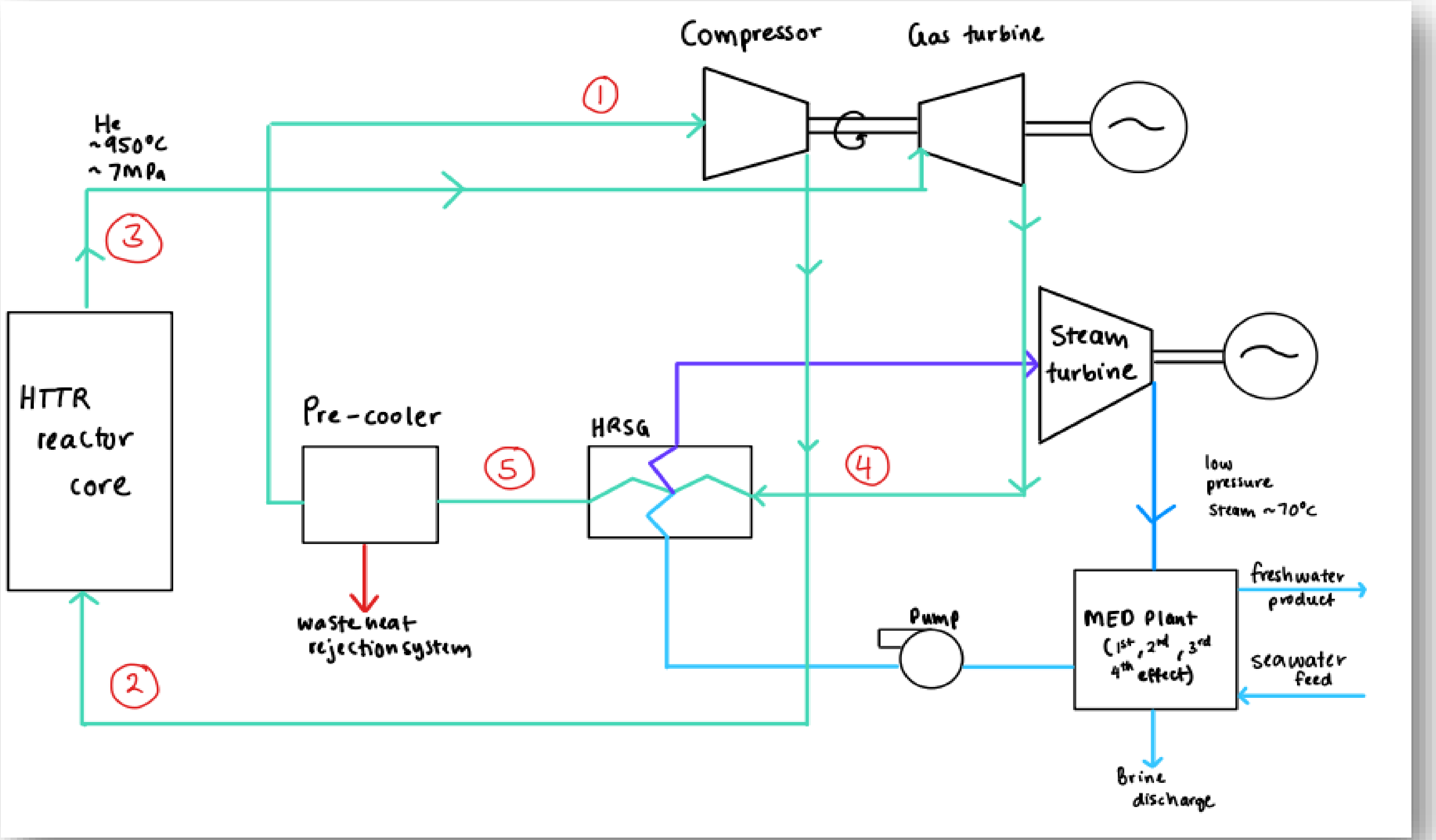
2. Literature Review
2.2. CoolProp Helium Fluid Properties Integration

Limitations of Simulink

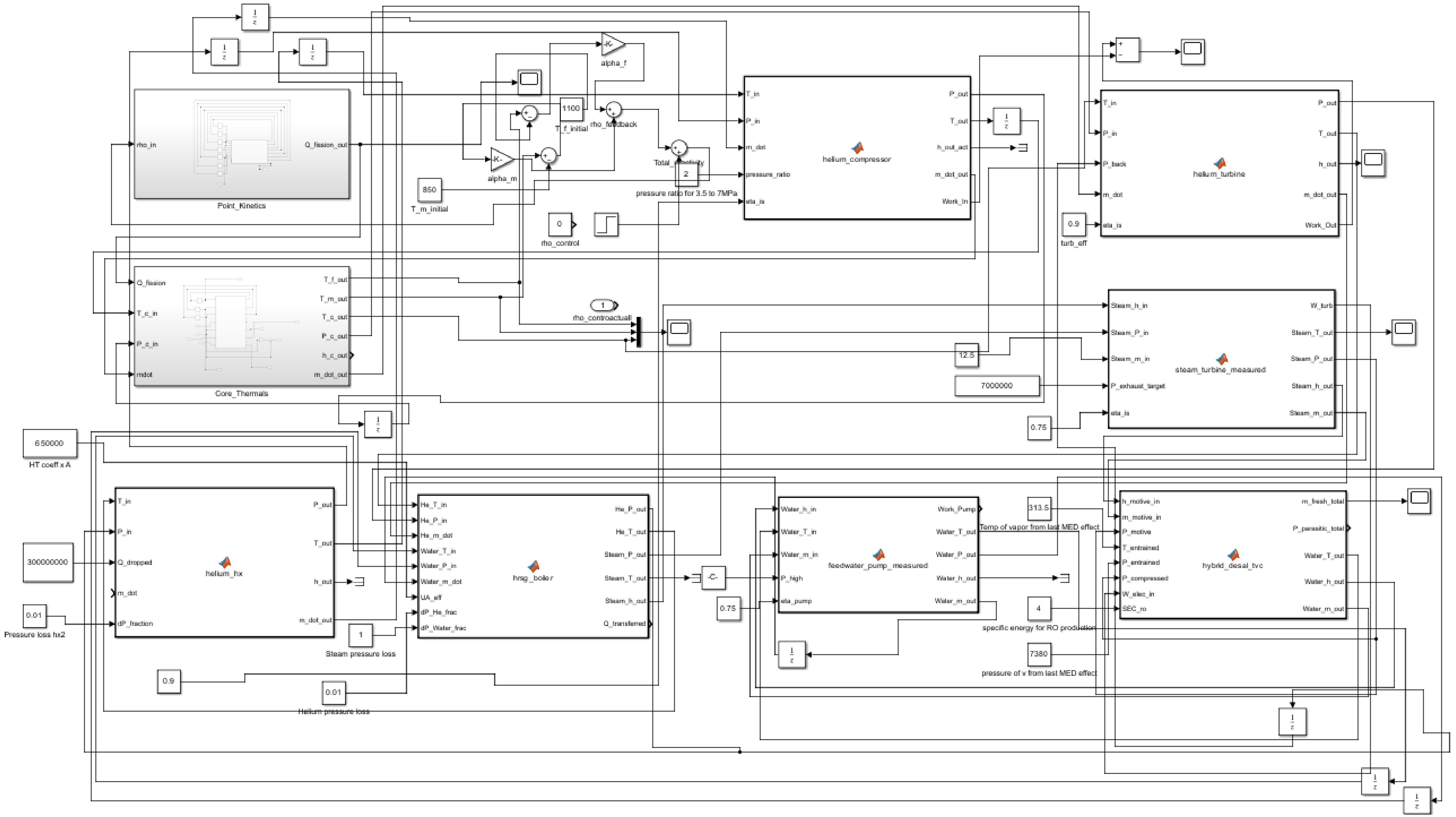




3. Complex Model Configuration
3.1. The First Simplified drawn-out model of the HTTR-Desalination



3.2. Final Simulink Model of HTTR-Desalination Complex System



4. Simulation Data and Fluid Properties

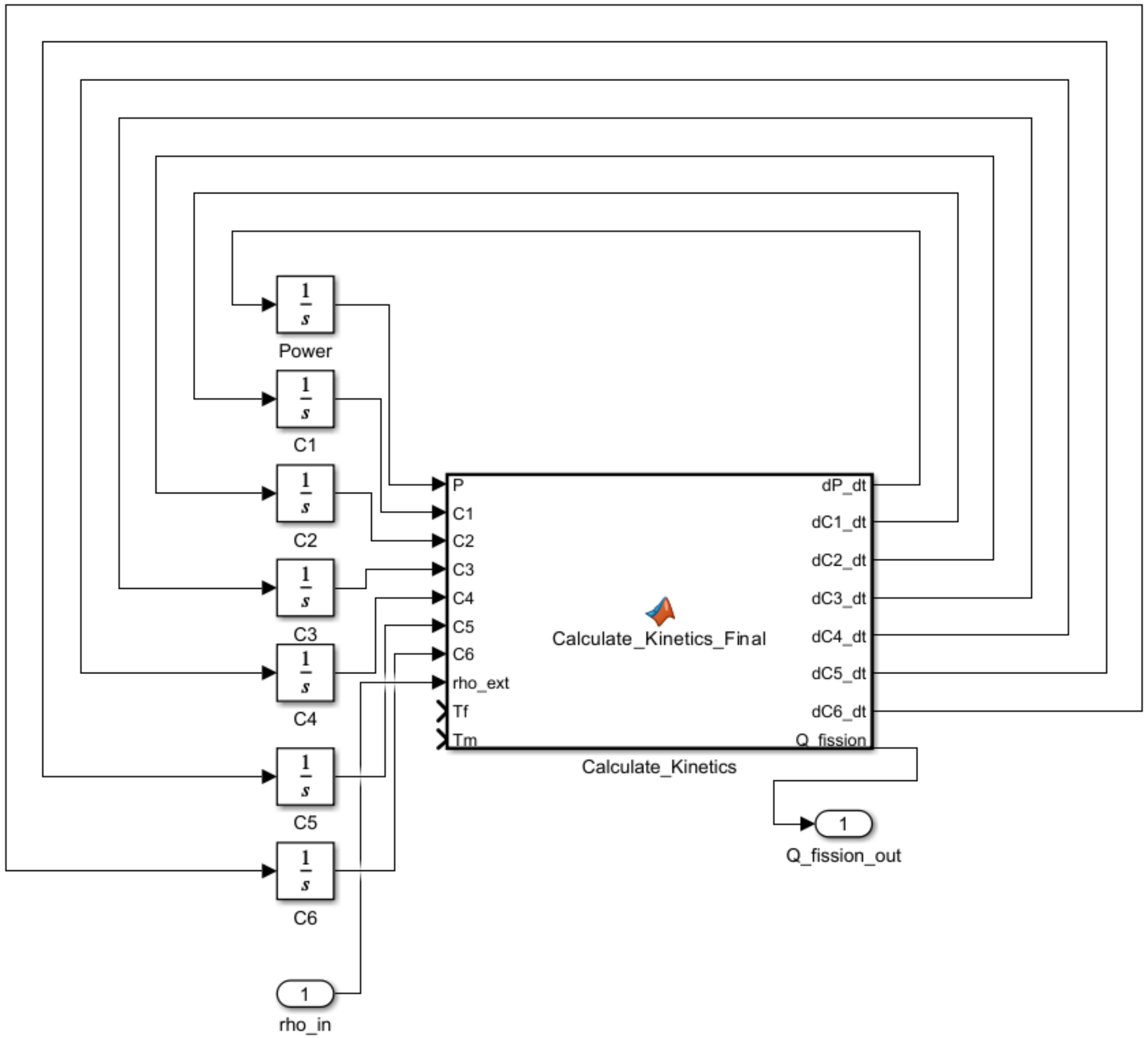
4.1. Kinetic Modelling Assumption

$$Reactor\ Power: \frac{dP(t)}{dt} = \left(\frac{\rho(t)-\beta}{\Lambda}\right) P(t) + \sum_{i=1}^{6} \lambda_i C_i(t)$$

$$Precursor\ Groups: \frac{dC_i(t)}{dt} = \left(\frac{\beta_i}{\Lambda}\right) P(t) - \lambda_i C_i(t)$$

Reactor Neutronics (Point Kinetics) Parameters

- $P(t)$: Normalised reactor power.
- $C_i(t)$: Normalised concentration for group i.
- $\rho(t)$: Total reactivity.
- β: Total delayed neutron fraction.
- Λ: Neutron lifetime (s).
- λ_i: Decay constant for group i (s^{-1}).
- $Q_{fission}$: Thermal power calculated by $P(t) \cdot Q_{nominal}$.



4. Simulation Data and Fluid Properties

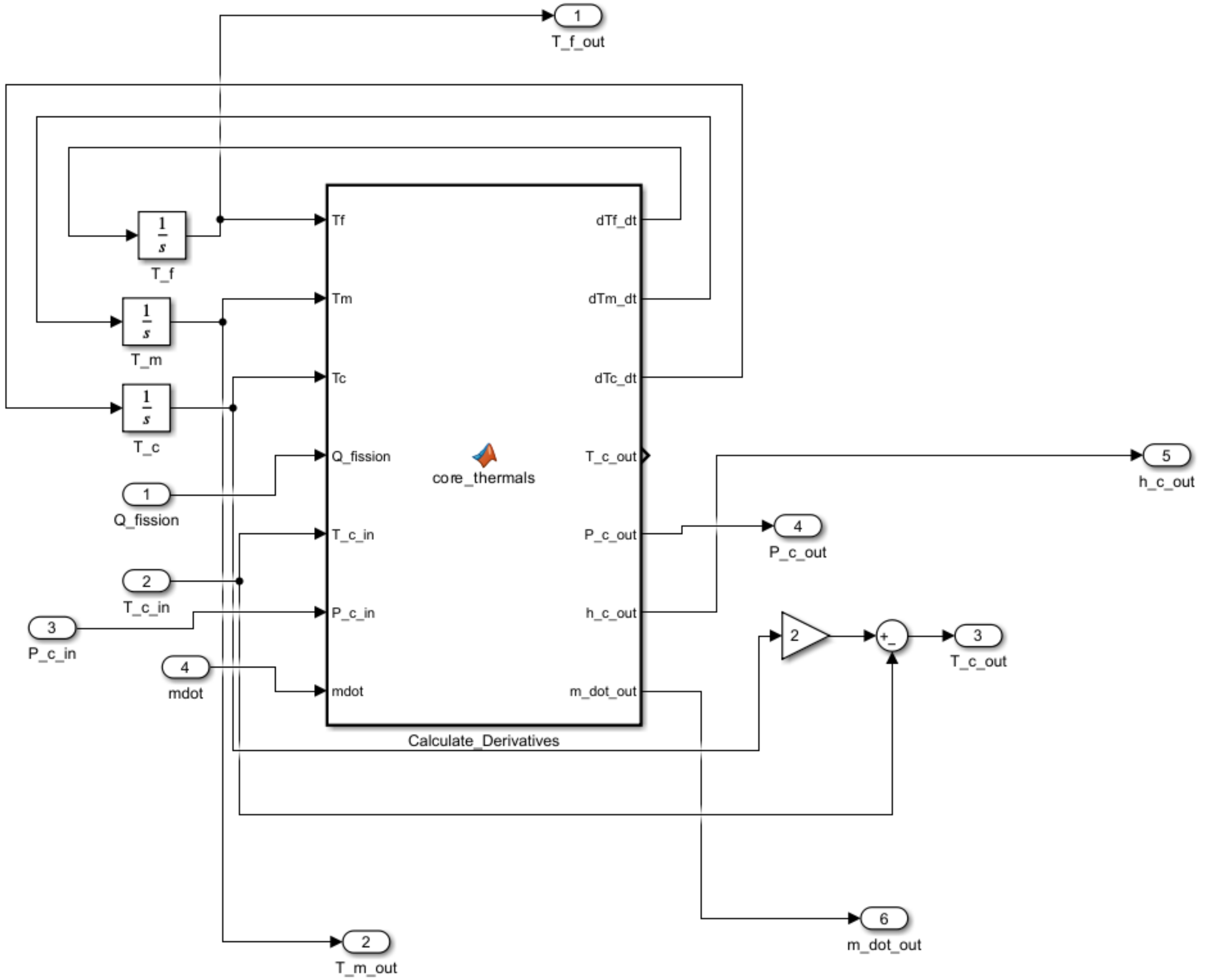
4.2. Thermal Modelling Assumption:

Fuel Temperature: $C_f \frac{dT_f}{dt} = Q_{fission} - h_{fm}(T_f - T_m) - h_{fc}(T_f - T_c)$

Moderator Temperature: $C_m \frac{dT_m}{dt} = h_{fm}(T_f - T_m) - h_{mc}(T_m - T_c)$

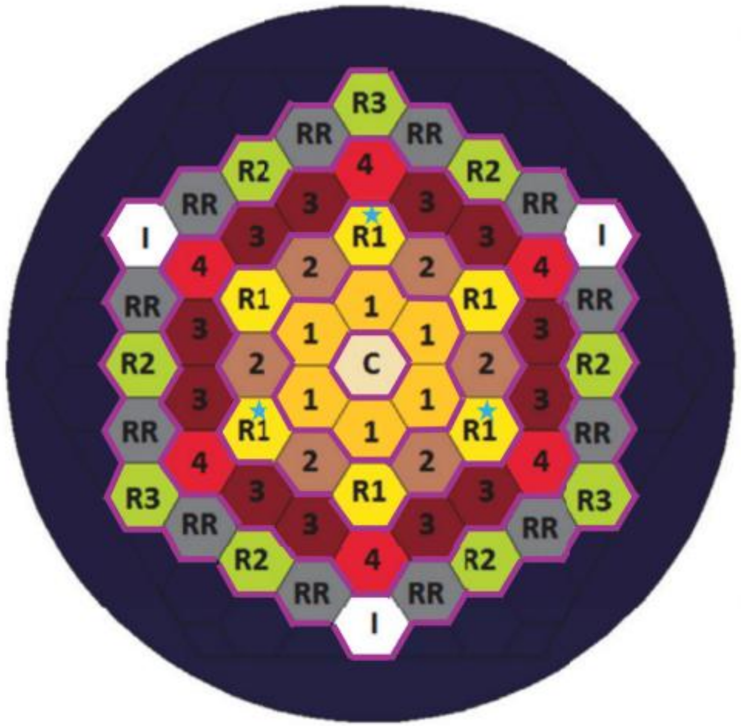
$Coolant\ Temperature\colon\ C_c \frac{dT_c}{dt} = h_{fc}(T_f - T_c) + h_{mc}(T_m - T_c) - 2\dot{m}C_p(T_c - T_{in})$

- T_f **(Fuel Temp):** Average temperature inside the TRISO fuel kernels (K).
- T_m **(Moderator Temp):** Temperature of the bulk graphite matrix (K).
- T_c **(Coolant Temp):** Bulk average Helium temperature inside the core (K).
- $Q_{fission}$**:** Total thermal power input (300 MWt Nominal).
- C_x **(Thermal Capacity):** The "Heat Mass" ($m \cdot C_p$) of each respective region.
- h_{xy} **(Heat Transfer Coeff.):** The physical efficiency of heat moving between nodes.
- $\dot{m}$ **(Mass Flow):** Primary Helium flow rate maintained at **105 kg/s**.
- T_{in} **(Inlet Temp):** Temperature of Helium returning from the primary loop ($350°\mathrm{C}$).



4.3. Pronghorn Model and Laminar Flow Assumption

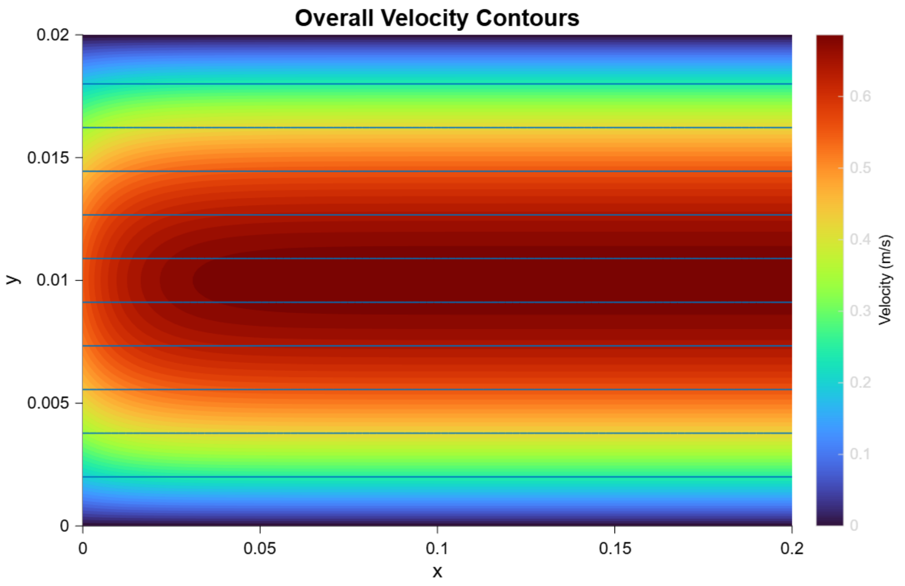
The model was coded and built using the pronghorn model and laminar flow



Continuity equation: $\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$

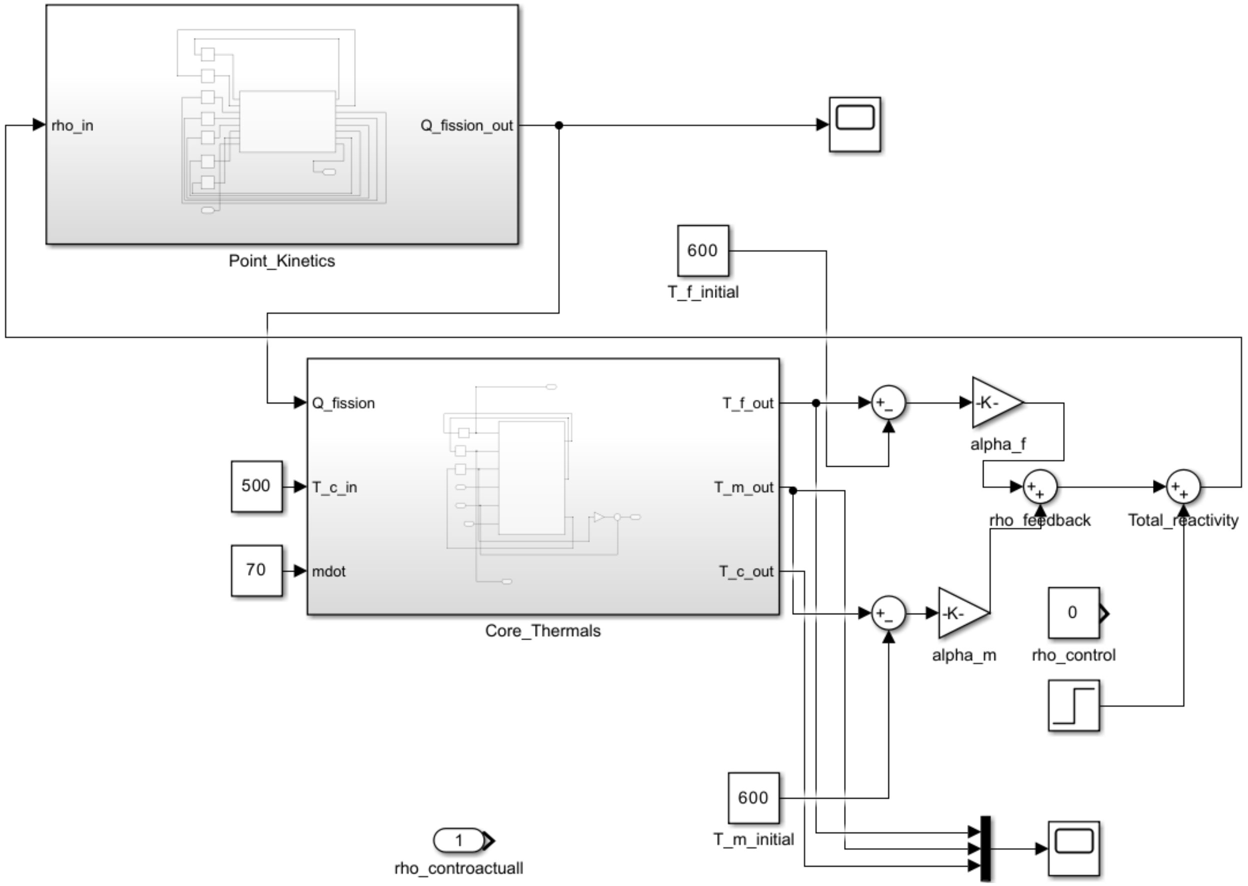
Momentum in x- direction: $u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial x} + \nu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right)$

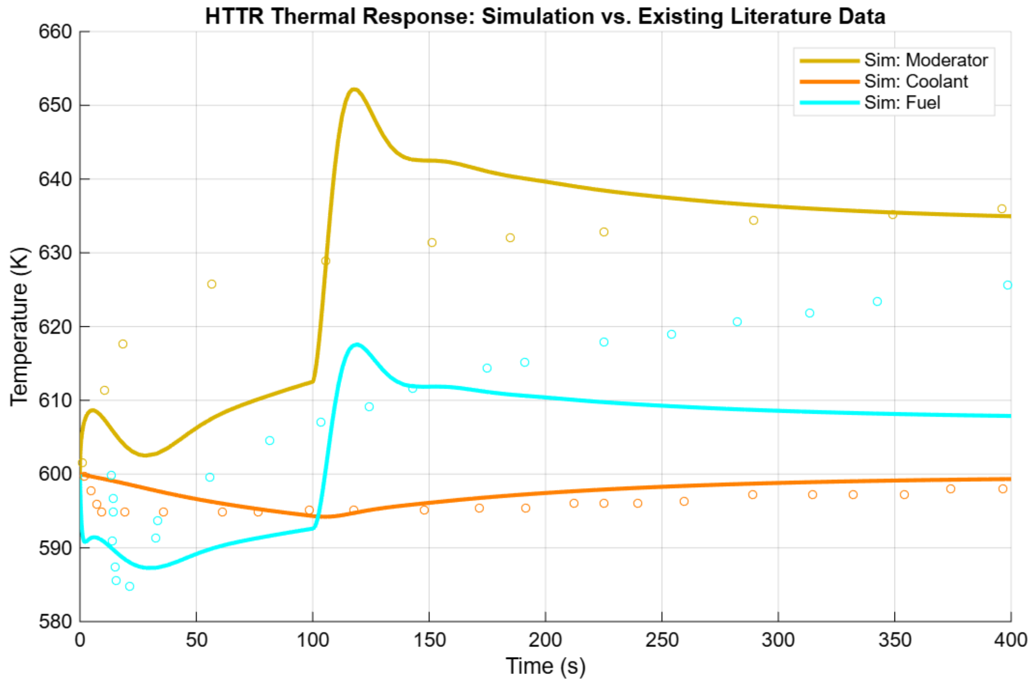
$Momentum\ in\ y -\ direction:\ u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{1}{\rho}\frac{\partial p}{\partial y} + \nu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right)$



$$A = 6 \times \frac{y^2}{tan(\pi/3)} \qquad 6 \times \frac{y^2}{tan(\pi/3)} = \pi r^2 \qquad r = \sqrt{\frac{6y^2}{\pi \cdot tan(\pi/3)}} \approx 1.05y$$

4.4.1 Thermal Core Fluids Comparison





4.4. Verification and Validation
4.4.2 Benchmark Figures of Merit (FOM) for the High Temperature Nuclear Reactor

FOM	Steady State Deviation	Time series (TS) or Scalar (S)	Validation?
Outlet Coolant Temperature	0.3%	TS	Yes
Moderator Temperature	~1.1%	TS	Yes
Average Fuel Block Temperature	~7.3%	TS	Partially
Inlet Coolant Temperature	<0.5%	S	Yes

The HTTR: 30 MWt vs. 300 MWt

4.4.3. Validity on assuming a 10x scale for commercial usage

We have scaled the parameters up to a **Commercial/Utility Scale (300 MWt)** for two specific reasons:

Population Demand: A 30 MWt reactor would only provide enough water for about 8,000 people. To serve our **target city of 80,000 people per day**, we need to roughly scale up to **300 MWt**

Economic Viability: The cash flow graph shows a 27% IRR. This high profitability usually only happens at the 300 MWt scale due to "economies of scale." A smaller **30 MWt plant has much higher costs per litre of water** produced.

This has been taken into account when designing our nuclear reactor in Simulink/MATLAB

```
% 4. DIFFERENTIAL EQUATIONS (Point Kinetics)
% ============================================================================
% Neutron Population (Power) Derivative
sum_lambda_C = sum(lambda .* [C1, C2, C3, C4, C5, C6]);
dP_dt = ((rho_total - beta_total) / Lambda) * P + sum_lambda_C;

% Precursor Group Derivatives
dC_vec = (beta_i ./ Lambda) .* P - lambda .* [C1, C2, C3, C4, C5, C6];
dC1_dt = dC_vec(1); dC2_dt = dC_vec(2); dC3_dt = dC_vec(3);
dC4_dt = dC_vec(4); dC5_dt = dC_vec(5); dC6_dt = dC_vec(6);

% ============================================================================
% 5. FINAL OUTPUT TO CORE THERMALS
% ============================================================================
Q_fission = P * Q_nominal; % Scaled Power in Watts

end
```

From our Kinetics Block, the Q_nominal we manage to achieve is 300MWt and is normalised with P(t)=1, which indicates a 300MWt energy transfer into the Thermal Block.

Furthermore, the mass flow rate of helium in the original HTTR is **10.5-12.3kg/s; hence, in our model, we are using 105kg/s**

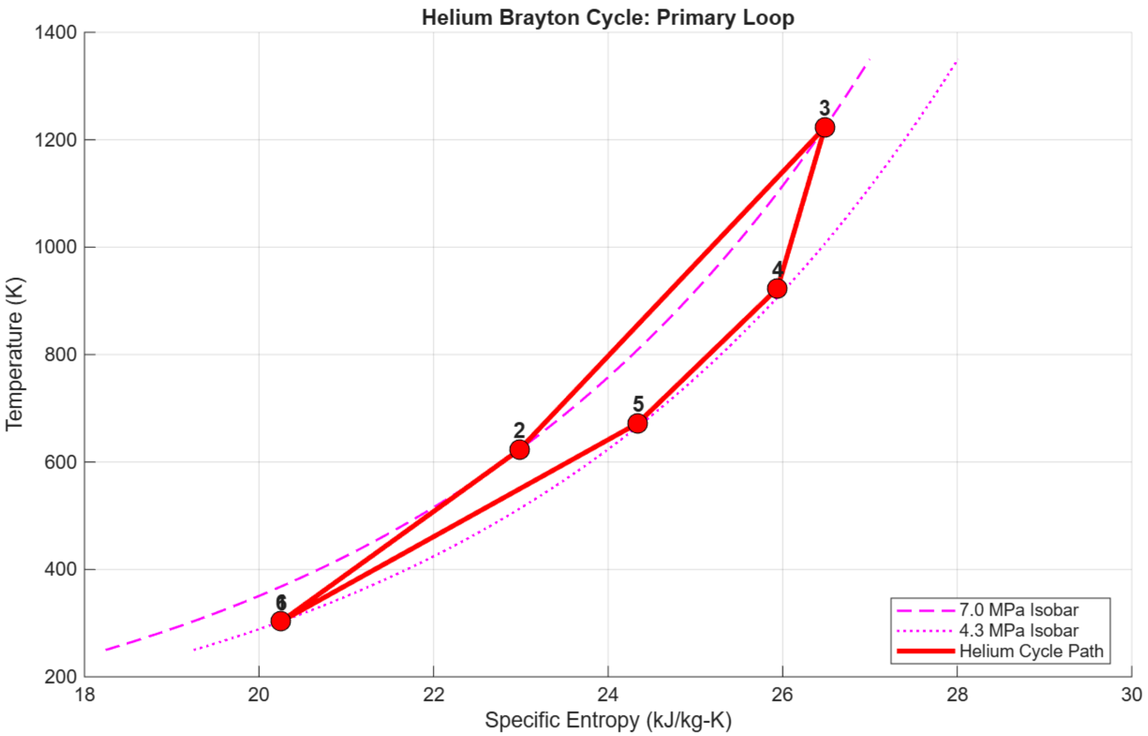
5. Thermodynamic Analysis of Power Extraction Cycles

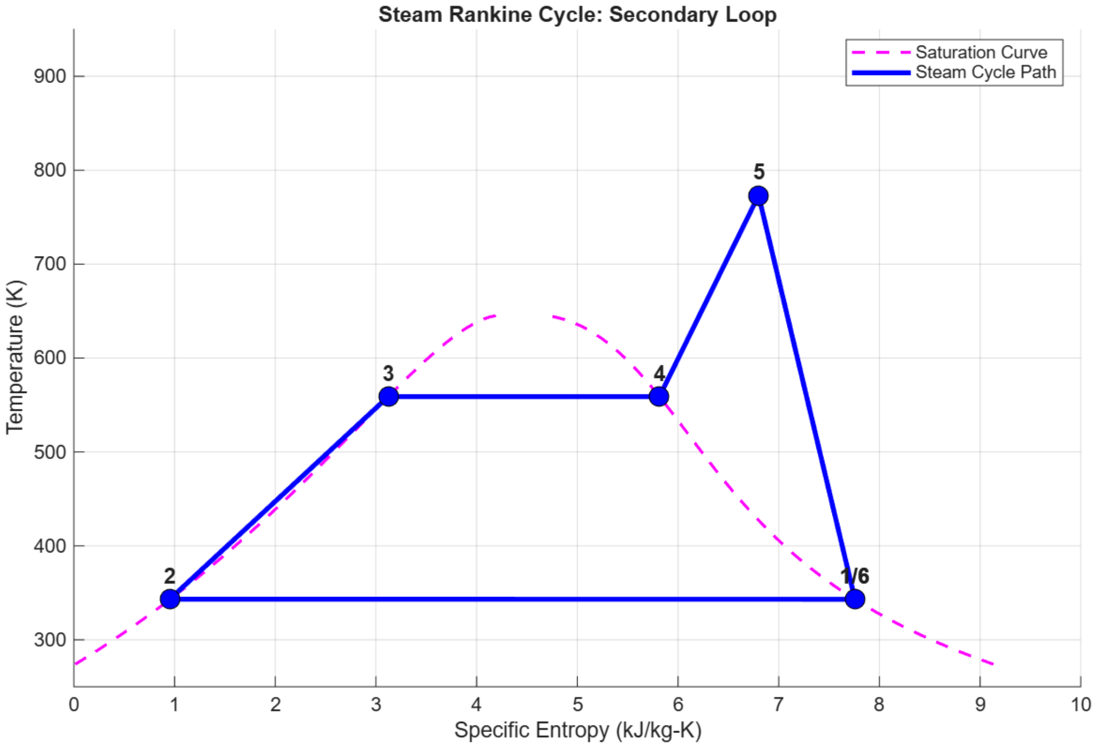
5.1. Target Parameters for our Power Extraction Cycles (Brayton and Rankine Cycle)

Brayton and Rankine Cycle (FOM) Parameters:

Loop	Pt.	Component	T (°C)	$\dot{m}$ (kg s^{-1})	P (bar)	Power/Heat (MW)
Primary (He)	1	Compressor inlet	30	105.0	43	−205.1
–	2	Compressor outlet	350	105.0	70	—
–	3	Reactor outlet	950	105.0	70	+300
–	4	Gas turbine exit	650	105.0	45	+147.4
–	5	HR system inlet	399	105.0	44	−195
–	6	HR system outlet	30	105.0	43	−47.3
Secondary (Water)	A	Feedwater pump out	70	12.5	70	−0.12
–	B	Steam turbine in	500	12.5	70	+195.0 (heat)
–	C	Steam turbine out	70	12.5	0.31	+12.1 (work)
–	D	Desalination in	70	12.5	0.31	+182.8 (heat)

5.2. Thermodynamic Analysis of Power Extraction Cycles





Primary Loop (Helium Brayton Cycle)

- **Total Reactor Thermal Input (Q_{in})**: $\mathbf{300}\ \mathrm{MWt}$
- **Gross Gas Turbine Output ($W_{turb,He}$)**: $\mathbf{147.4}\ \mathrm{MW}$
- **Compressor Work Required ($W_{comp,He}$)**: $-\mathbf{205.1}\ \mathrm{MW}$
- **Thermal Energy Transferred to Secondary Loop**: $\mathbf{195.0}\ \mathrm{MWt}$ (via HRSG/Boiler)
- **System Balancing**: The Helium turbine is sized primarily to drive the high-pressure compressor, maintaining the $950^{\circ}\mathrm{C}$ cycle.

Secondary Loop (Steam Rankine Cycle)

- **Thermal Input from Primary Loop**: $\mathbf{195.0}\ \mathrm{MWt}$
- **Gross Steam Turbine Output ($W_{turb,St}$)**: $\mathbf{12.1}\ \mathrm{MW}$ (Optimized for $70^{\circ}\mathrm{C}$ exhaust)
- **Feedwater Pump Work ($W_{pump,St}$)**: $-\mathbf{0.12}\ \mathrm{MW}$
- **Net Electrical Output**: $\mathbf{11.98}\ \mathrm{MWe}$
- **Energy Available for Desalination (Q_{desal})**: $\sim\mathbf{182.8}\ \mathrm{MWt}$
  - This "Recycled Heat" is the driver for the $12,000\ \mathrm{m}^3/\mathrm{day}$ production.

5.3 Benefits of using a combined power extraction cycle

Efficiency of System + Carbon Dioxide Emissions

Lower Carbon Dioxide emissions per unit of electricity generated

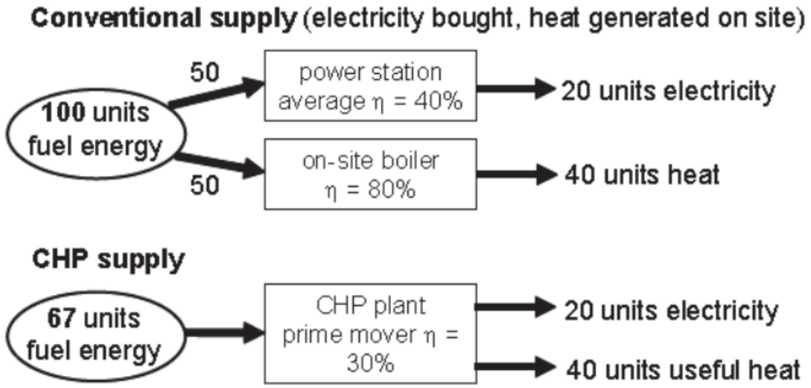


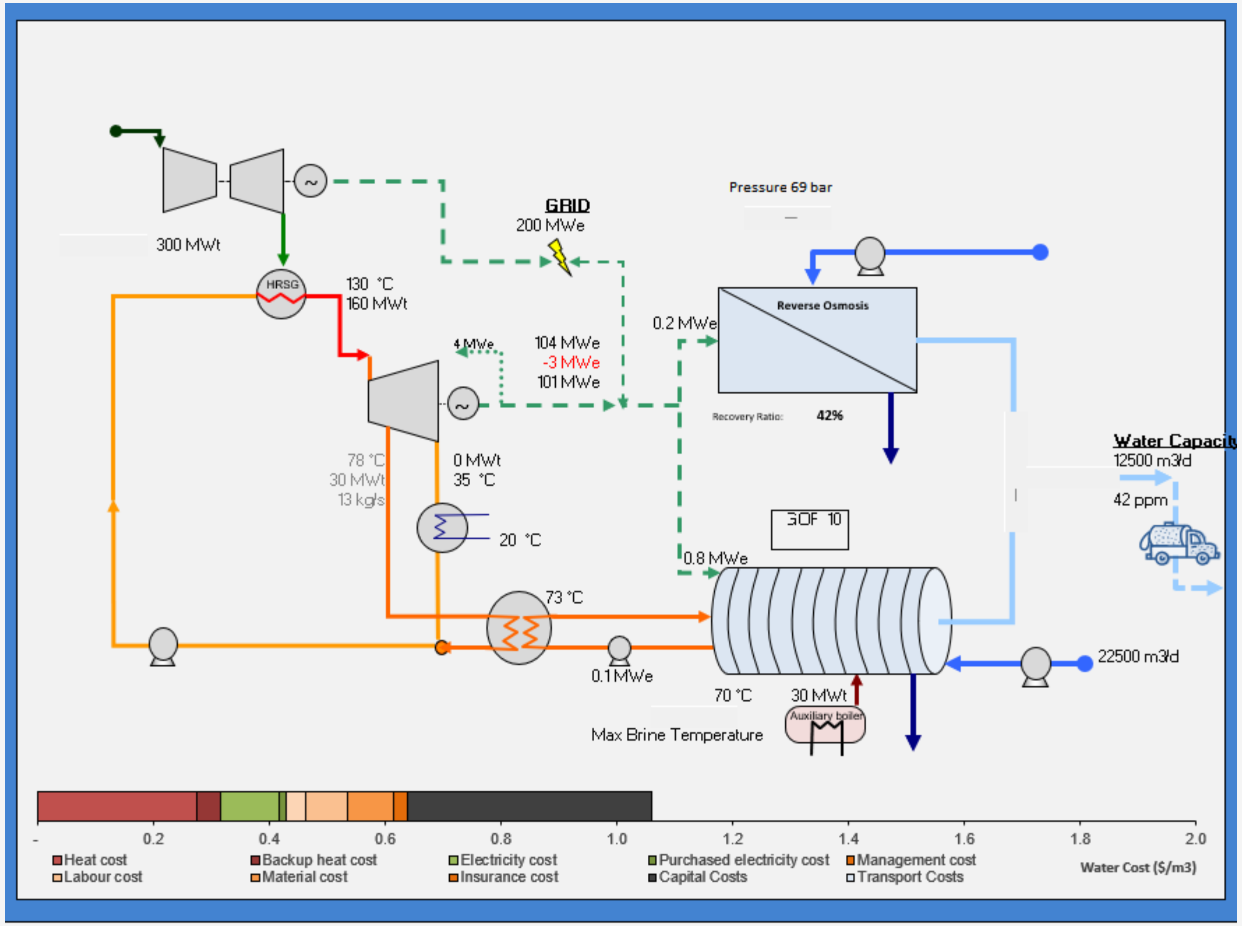
Figure 3.40: An example of comparing a conventional supply and a CHP supply.

$$\frac{electricity + useful\ heat\ output}{energy\ in\ fuel} = \text{energy utilization factor } (EUF)$$

Parameter	Helium Brayton	Steam Rankine	Combined
T_{max}	1220 K	773 K	1220 K
T_{min}	923 K	343 K	343 K
Carnot efficiency	24.5 %	55.6 %	72 %

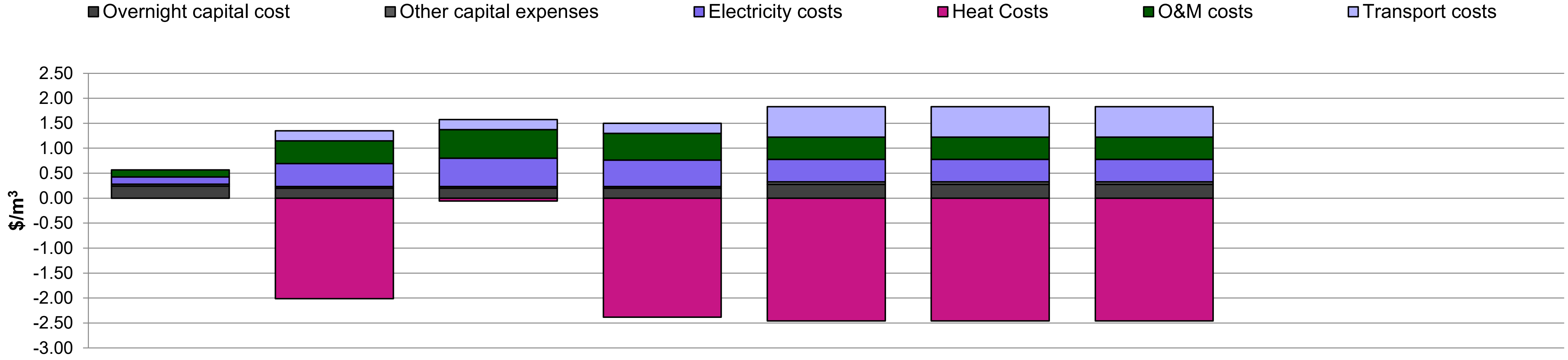
6. Economic Evaluation of Coupled Nuclear Reactor and Desalination Plant

6.1 Identical Model Configuration in DEEP and Desalination Plant Cos



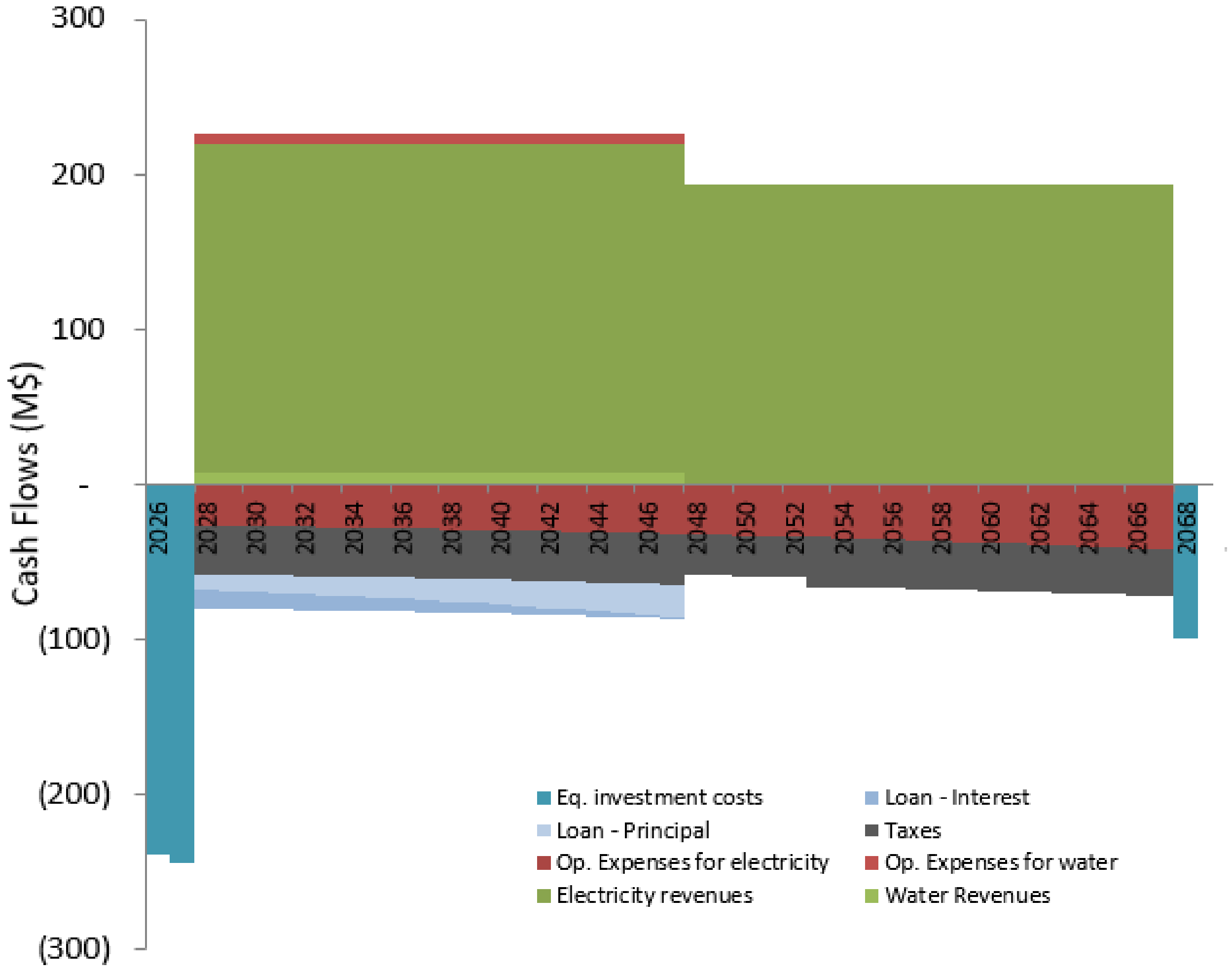
6.2 DEEP economic evaluation of various complex configurations

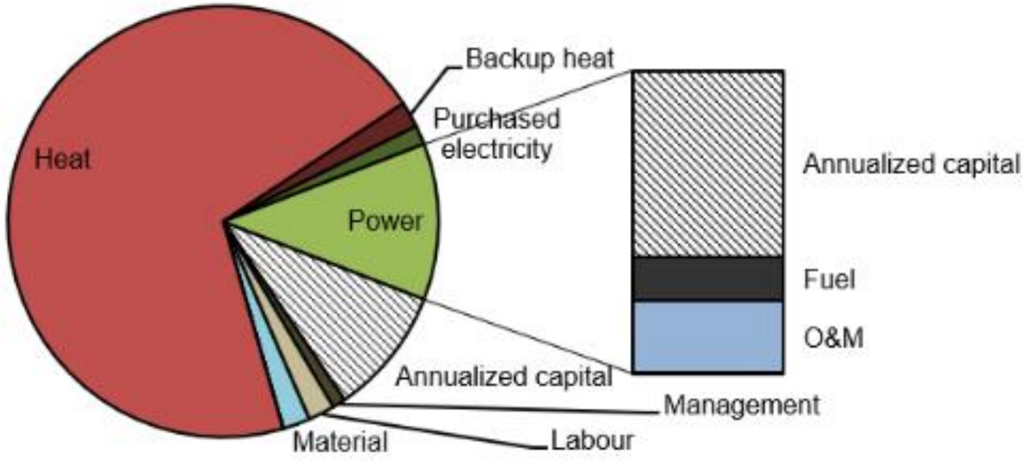
Analysis of why our complex is most profitable

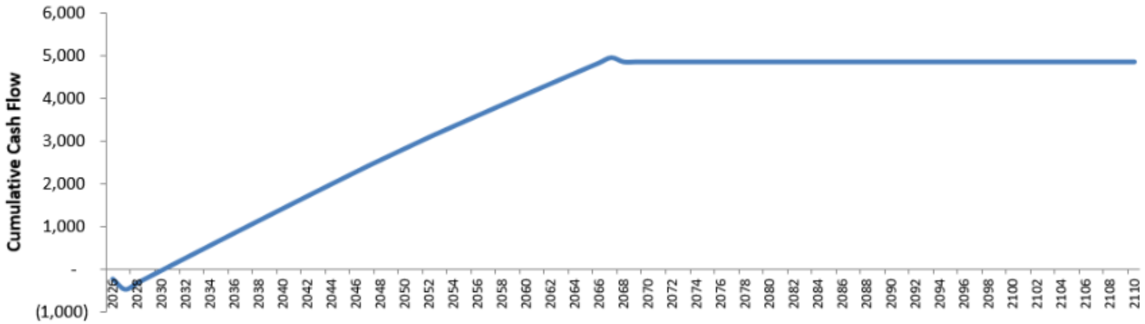


Current model run: Outputs Summary			Comb + Hyb	Comb + MED	Gas + MED	Steam + MED	Comb + Hyb	Comb + Hyb	Comb + Hyb
Levelized Capital Costs	0.33	$/m3	0.28	**0.23**	**0.23**	**0.23**	**0.33**	**0.33**	**0.33**
Base plant overnight EPC	0.27	$/m3	0.24	**0.20**	**0.20**	**0.20**	**0.27**	**0.27**	**0.27**
Other	0.05	$/m3	0.04	**0.04**	**0.04**	**0.04**	**0.05**	**0.05**	**0.05**
Levelized operating costs	-1.21	$/m3	0.38	**-1.21**	**0.86**	**-1.50**	**-1.21**	**-1.21**	**-1.21**
Heat	-2.46	$/m3	0.00	**-2.01**	**-0.06**	**-2.38**	**-2.46**	**-2.46**	**-2.46**
Electricity	0.45	$/m3	0.14	**0.46**	**0.57**	**0.53**	**0.45**	**0.45**	**0.45**
O&M	0.19	$/m3	0.24	**0.14**	**0.14**	**0.14**	**0.19**	**0.19**	**0.19**
Transport	0.61	$/m3	0.00	**0.20**	**0.20**	**0.20**	**0.61**	**0.61**	**0.61**
Lifecycle Emissions	47	Mtn/yr	31	31	**15**	**14**	**47**	**47**	**47**
Thermal Utilization	60%		67%	**37%**	**-29%**	**-2%**	**60%**	**60%**	**60%**
Power lost	-25.2	MWe	0	**-62**	0	**-62**	**-25**	**-25**	**-25**
Power used for desal.	4	MWe	4	**13**	**13**	**13**	**4**	**4**	**4**
Power cost	50.8	$/MWh	42.0	42.0	**53.8**	**49.9**	**50.8**	**50.8**	**50.8**

6.3. Economics of Overall Power Plant
Graph of Cash Flows over 20 years time







7.1. Appendix 1: Calculation of estimated water production per day

Data:

$$P_{\text{thermal}} = 300\ \text{MW}_{\text{t}} = 300\ \text{MJ/s}$$
$$t_{\text{day}} = 86{,}400\ \text{s/day}$$
$$\dot{\text{m}}_{\text{steam}} = 12.5\ \text{kg/s}$$
$$\text{GOR} = 10 - 12\ (\text{e.g.}\ 11.1)$$

Calculation:

Step 1: Total Steam Input per Day

$$\dot{\text{m}}_{\text{steam,day}} = \dot{\text{m}}_{\text{steam}} \times t_{\text{day}} = 12.5 \times 86{,}400 = 1{,}080{,}000\ \text{kg/day}$$

Step 2: Theoretical Water Output (GOR = 11.1)

$$V_{\text{water}} = \dot{\text{m}}_{\text{steam,day}} \times \text{GOR} = 1{,}080{,}000 \times 11.1 = 11{,}988{,}000\ \text{kg/day}$$

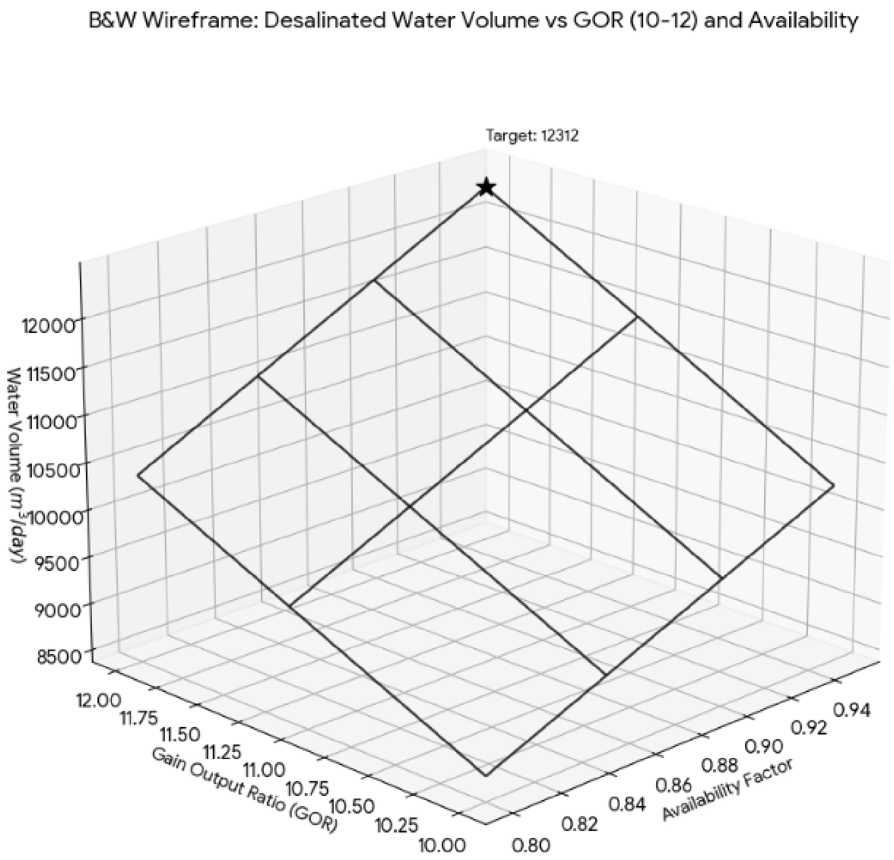
Step 3: Volumetric Conversion

$$V_{\text{vol}} = \frac{V_{\text{water}}}{\rho_{\text{water}}} = \frac{11{,}988{,}000\ \text{kg/day}}{1{,}000\ \text{kg}/m^3} \approx 12{,}000\ m^3/\text{day}$$

Step 4: Applying Availability Factor (95%)

$$V_{\text{actual}} = V_{\text{vol}} \times \eta_{\text{avail}} = 12{,}000 \times 0.95 = \mathbf{11{,}400}\ \boldsymbol{m^3}/\mathbf{day}$$

Supply Chain: Nuclear Thermal Energy → Zero Emission Water → Zero Emission Power



7. Calculation of Net CO2 saved over the duration of our proje

7.2. Appendix 2: Sample calculation of estimated CO2 reduced per year

1. Calculation for Desalination (Water)

$\text{Production basis} = 12{,}000 \times 0.95 \times 36 = 4{,}161{,}000\ \text{m}^3/\text{year}$

$\text{Emission Factor (Avoided)} = 5\ \text{kg CO}_2/\text{m}^3$
$(footprint\ of\ typical\ fossil - fuel\ thermal\ desalination)$

Calculation :
$4{,}161{,}000 \times 5 = \mathbf{20,805\ tonnes\ CO_2/year}$

2. Calculation for Electricity (Power)

$\text{Capacity} = 40\ \text{MW}_\text{e}$

$\text{Annual generation} = 40 \times 0.95 \times 8{,}760 = 332{,}880\ \text{MWh/year}$

Emission Factor = $0.74\ \text{CO}_2/\text{MWh}$

Calculation :
$332{,}880 \times 0.74 = \mathbf{246,331\ tonnes\ CO_2/year}$

Water Produced (m³/day)	85% Avail	90% Avail	95% Avail
10,000	15,512	16,425	17,337
12,000	18,615	19,710	20,805
14,000	21,717	22,995	24,272

$\mathbf{Total\ Environmental\ Impact} = 20{,}805 + 246{,}331 = \mathbf{267,136\ tonnes\ CO_2/year}$

During the Project lifespan (20–30 years) = $\mathbf{8\ Million\ tonnes\ of\ CO_2 Avoided}$

8. Reactor and Model Safety

8.1. Environmental Factors + Process Dynamics and Control

Synergistic Cooling: Cogeneration reduces thermal pollution by "recycling" 183 MWt of heat into water production instead of the ocean.

Marine Biodiversity Protection: Brine management protocols ensure local salinity and temperature remain within ecological safety limits.

Air Quality Preservation: Zero-emission operation eliminates the "hidden" environmental health costs of fossil-fuel combustion.

Geological Stability: The prismatic core design is inherently resistant to seismic activity, a critical factor for coastal desalination sites.

1) Process Control- Uses Proportional-Integral-Derivative (PID) controllers to manage thermal inertia, ensuring that even during a turbine trip, the 950°C core temperature remains within the safe structural limits of the graphite matrix.

2) Feedback and Master Loops - Automatic shutdown at 90% prompt criticality to avoid meltdowns

3) Variable pump rate to control He flow rate using control valves -Dynamic mass flow control (adjusting the **105 kg/s** helium flow) allows the plant to follow demand load without thermal-shocking the Heat Recovery Steam Generator (HRSG)

4) The secondary loop acts as a physical 'clean' barrier; by using an intermediate heat exchanger, the water for **80,000 people** is never in direct contact with the primary helium coolant, ensuring zero risk of cross-contamination

8. Reactor and Model Safety
8.2. Project Management and Cross-Functionality

Collaborative Framework: Utilised a shared development environment (MATLAB folder, TEAMS, WhatsApp) to ensure cross-functionality between the Neutronics (Subtask 1) and Heat Rejection (Subtask 2) sub-teams.

Version Control & Quality: Adhered to **PEP8** standards for code readability and maintained a "Master State Table" to ensure all thermodynamic parameters were synced across models.

Technical Risk Capture:
Risk: Excessive "Pinch Point" in the HRSG leading to physical infeasibility.
Mitigation: Iterative pressure adjustments (optimising at 70 bar) to maintain a >30°C temperature margin.

Strategic Planning: Implemented a phased development approach—moving from steady-state verification (Phase 1) to dynamic transient analysis (Phase 2).

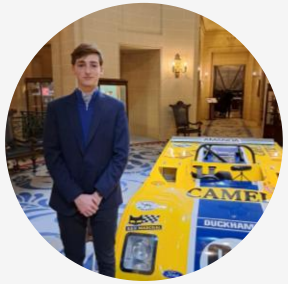
PART II: HR Contents

Introduction

Objectives + Team Structure

Aim	Design a novel water-independent heat rejection (HR) system compatible with High Temperature Gas Reactor (HTGR) plants
Design Concepts	**Previous:** Liebig condensers, pipe conduction **Current:** Shell-and-tube HE → Porous cube + Thermosyphon → Air Cooled Condenser
Challenges	Changing parameters, scheduling conflicts, unlucky accidents
Conclusion	Parasitic load of 21.5 kW per MW_{th} rejected is of industry standard; reduced **footprint** makes the concept promising for further development

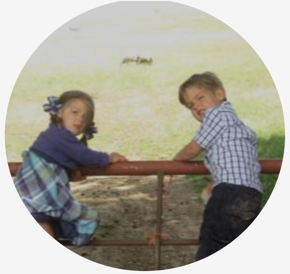
Oscar [y2 MechEng]
→ Project Manager
→ Secretary
→ Quality Assurance Lead
→ Design Lead



Ollie [y1 MechEng]
→ CAD Lead
→ Design Engineer

Heidi [y1 MechEng]
→ Research Lead
→ Design Engineer

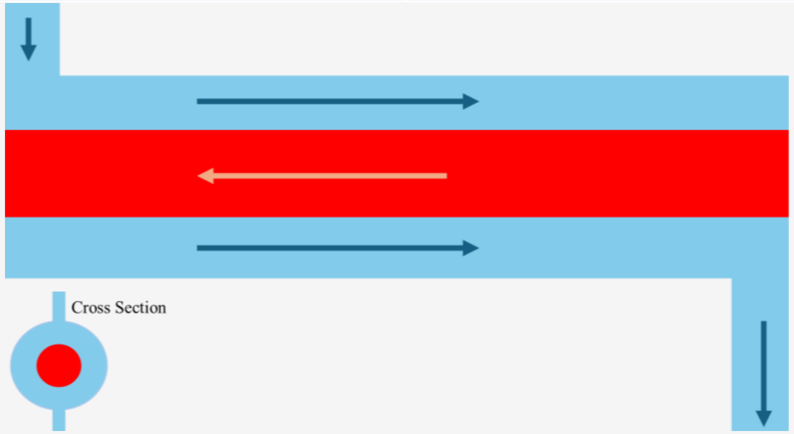
Muhammad [y2 EEE]
→ Electronics Analyst
→ Technical co-lead

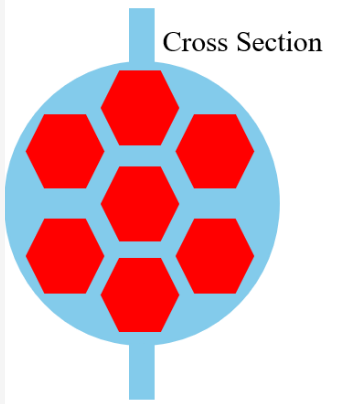


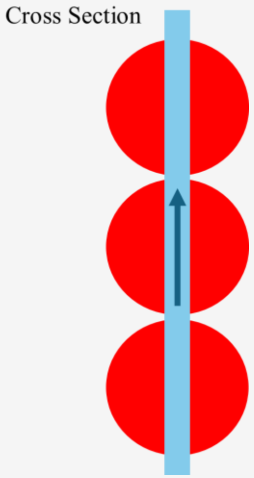
Luca [y2 ChemEng]
→ Technical co-lead

Literature Review

Design Idea	For	Against	Conclusion
Pipe conduction 1-3	Potential for maximisation for little added cost	Air conduction not sufficient for temperature change required.	Not suitable
Liebig 1	Promise due to more direct contact	Efficient at small scale, adjustments needed when larger	Not suitable
Liebig 2	Even more direct contact	More electricity to pumping.	Not suitable
Liebig 3	Large surface area and compacting fact to minimise size.	Difficult to manufacture and still not efficient enough	Not suitable
Porous Cube	*Volumetric condensation:* a 3D method to distribute heat evenly throughout the flow.	Difficult to quantify- porous density, permeability, etc. Pressure drop significant.	Suitable



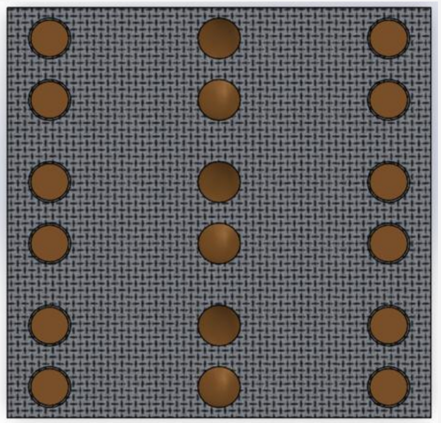


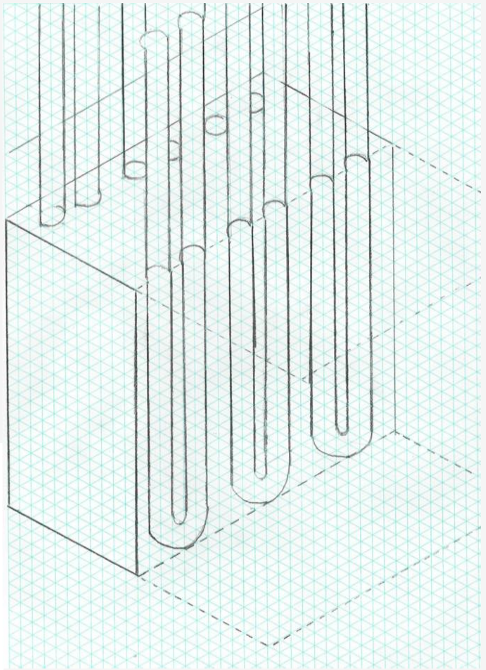


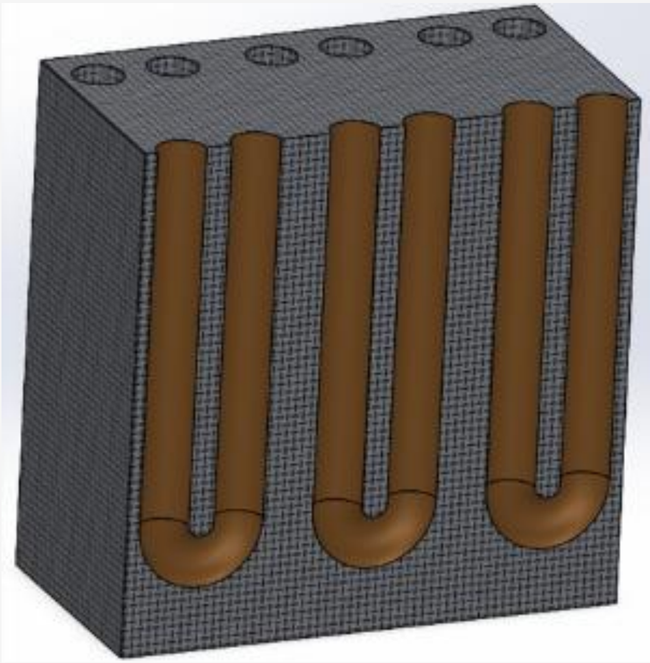
Design Justification
Pugh Matrix

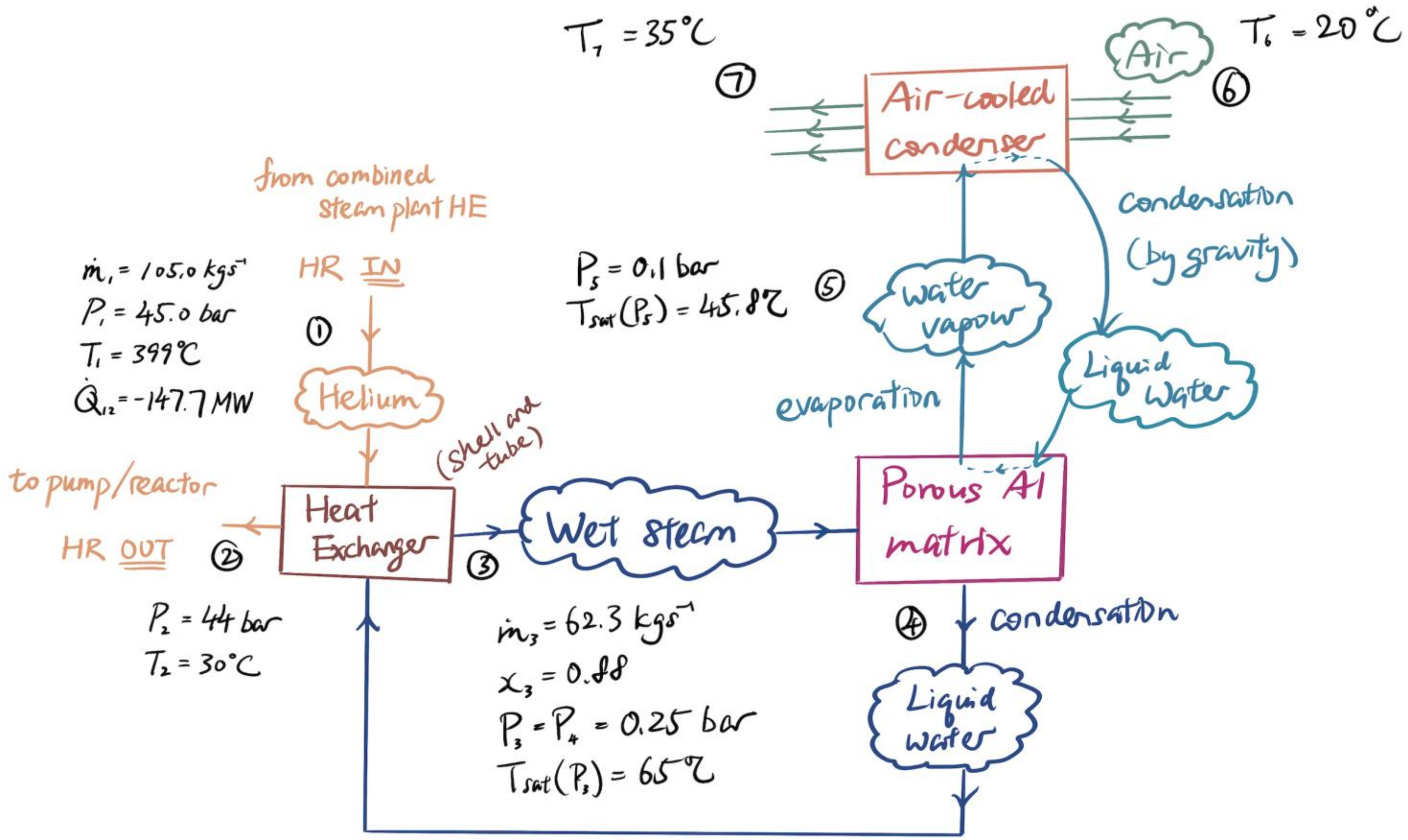
Criteria	Weighting	Air Condenser	Double-Pipe HE	Shell & Pipe HE (Liebig Condenser)	Porous Cube
Parasitic load	0.25	2	4	3	4
Heat transfer coefficient	0.25	3	2	4	5
Footprint	0.20	2	3	2	5
Safety	0.15	4	4	5	3
Maintenance	0.10	3	4	3	2
Cost	0.05	3	4	3	2
Total score	**1.00**	**2.70**	**3.00**	**3.35**	**4.00**

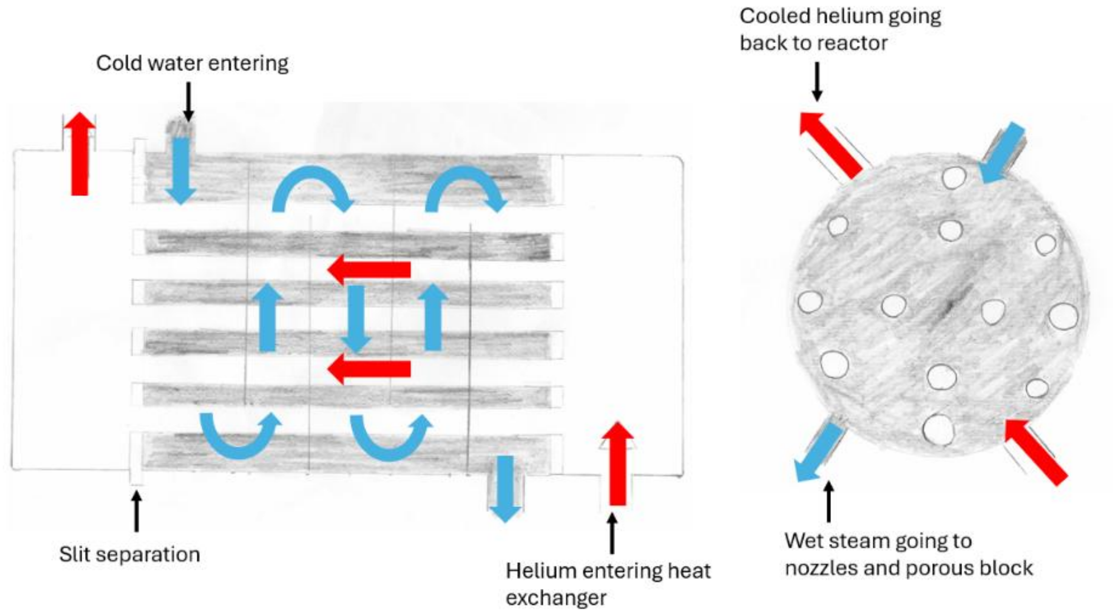
Design Justification
Final Design





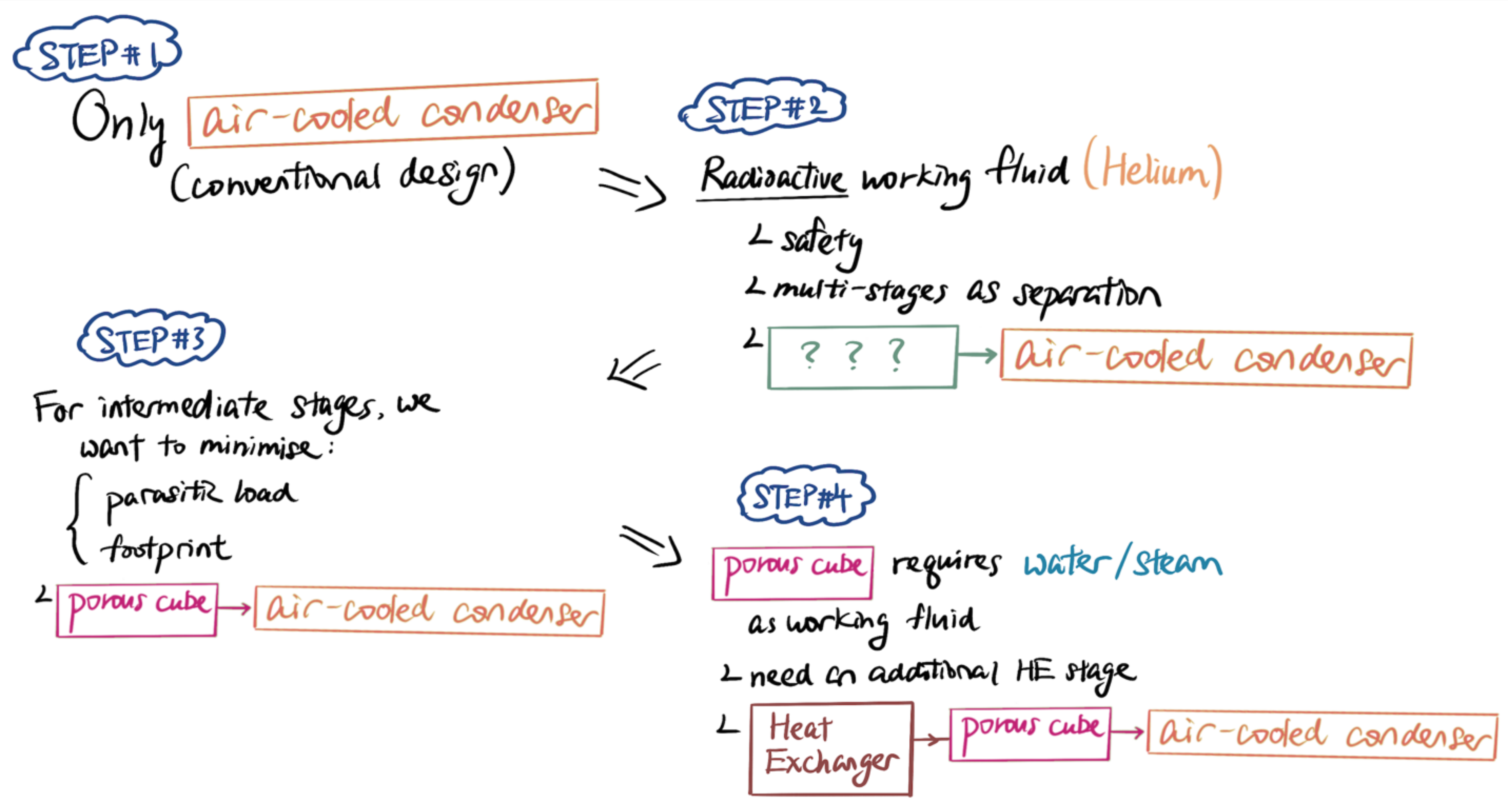






Design Justification

Down-selection and Decision-making Process



The Porous Metal as a Design – An Overview

Concept Overview:

- Steam condenses in porous metal receiver.
- Heat transported by thermosyphons.
- Rejected to air via dry-cooler modules.

Why This Architecture:

- Extremely high condensation surface area.
- Passive two-phase heat transport.

Scalability:

- System scales with thermosyphon count.

Economic Drivers

- Major capital costs: thermosyphon & air-cooler modules.
- Operating cost dominated by fan power.
- Feasibility depends on per-pipe heat capacity.

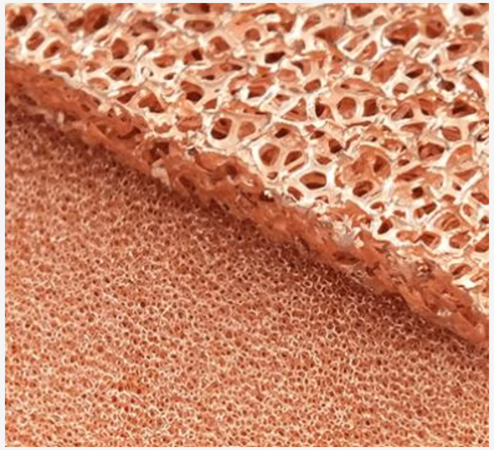
Engineering Challenges:

- Thermosyphon transport limits.
- Condensate drainage in porous receiver.

Further Validation Required:

- Thermosyphon performance at operating condition.
- Long-term durability of porous receiver.

Kelvion RF-NC101E4H dry cooler



Recemat copper foam (e.g. Cu-1116 grade)

Advanced Cooling Technologies (ACT) Thermosyphons

The Porous Metal as a Design – An Overview
Parasitic Loads

Condensate Loop Pump:
Modelled as a simple pipe route with Darcy–Weisbach friction losses.

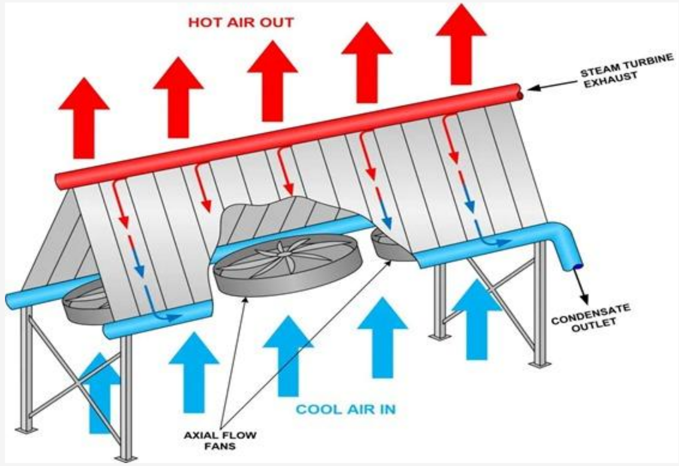
Assumptions:
- 6 long-radius elbows
- 2 gate valves fully open
- 1 swing check valve
- Length = 30 m
- velocity = 3 m/s
- Pipe roughness = 0.045 mm
- Pump efficiency = 0.75

$$P_{\text{pump}} = \frac{\rho g \dot{V} H}{\eta}$$

3.51 kW

Dry coolers:
Heat rejection modelled using manufacturer dry-cooler modules and air-side energy balance.

Assumptions:
- Surface Area = 197.1 m^2
- Volumetric Airflow Rate = 5.39 m^3/s
- Dimensions = 2220×1610×1350 mm (L × W × H)



Air-cooled condenser schematic illustrating airflow and condensation (fxstageadmin, 2020)

$$P_{\text{fan}} = \frac{\Delta P_{\text{air}} \times \dot{V}_{\text{total}}}{\eta_{\text{fan}}}$$

3.17 MW

Model Results & Post-Processing

Figures of Merit

Table showing baseline performance of the model.

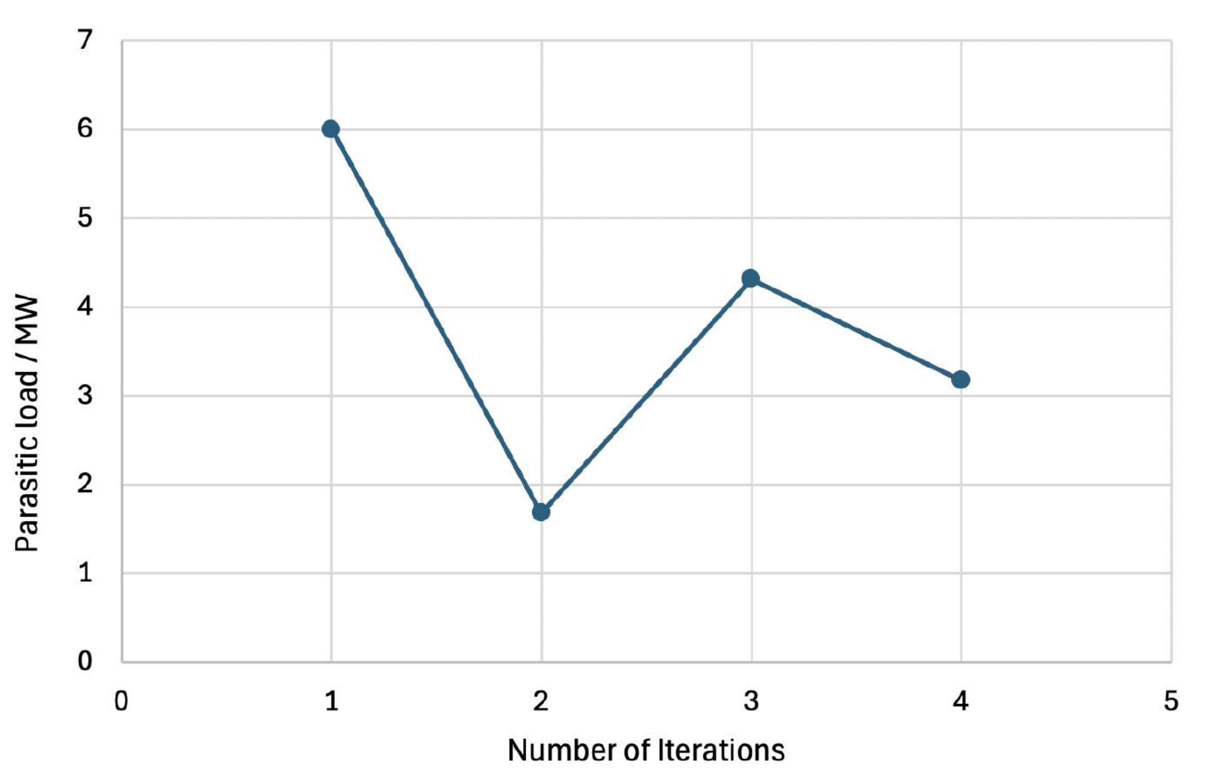
Temperature (°C)	Heat rejected (MWth)	Total Parasitic Load (MW)
20 °C	147.7	3.17

Table showing Figures of Merit of the model.

Parasitic load per MWth rejected (kW)	Number of modules required	Footprint (m^2)
21.5	1700	6021.4

Model Results & Post-Processing

Iterations with HTGR



Parasitic load against number of iteration with HTGR subteam.

Iteration #1
- literature review
- estimated 3% of ~200MW heat for air-cooled condensers

Iteration #2
- rejected 78MW heat from combined steam plant

Iteration #3
- returned to H_2 gas (Q = 205MW)
- pump power considered

Iteration #4
- utilised the updated 147.7MW value from HTGR after parameter exchange.

Discussion

Potential Improvements – Radiative Cooling Panels

$$q''_{total} = q''_{rad}(T_s) + q''_a(T_a) + q''_{solar} + q''_{conv}$$

Table X: Summary of best, worst and typical day case of radiative cooling panels.

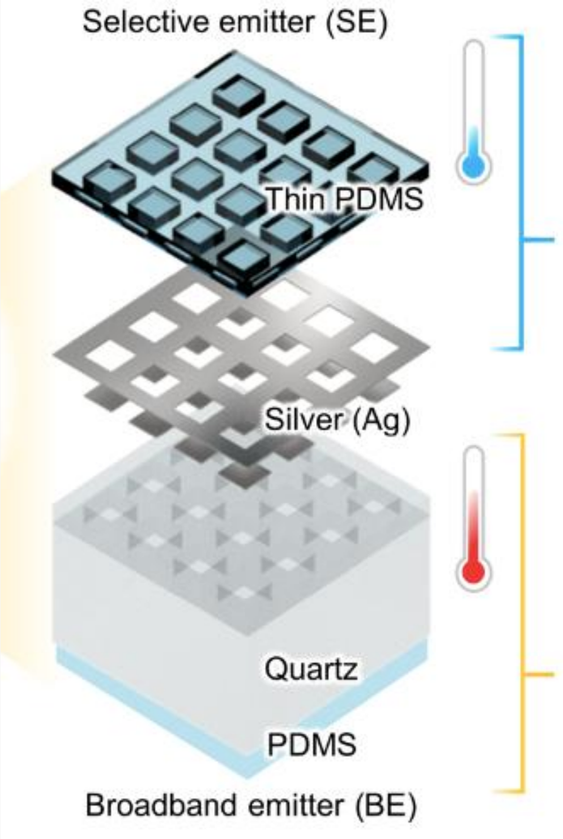
$[Wm^{-2}]$	Best case	Worse case	Typical day case
q''_{rad}	Unchanged; 629.026	Unchanged; 629.026	Unchanged; 629.026
q''_a	In January; 218.92	In July; 369.72	In warm summer; ~350
q''_{solar}	At night; 0	During daytime; 96.4	Whole day (avg); 48.2
q''_{conv}	Windy; 1536	No wind; 0	Min. avg wind; 774
q''_{total}	1947	162.9	1004.8

Good news – There is no theoretical parasitic load!

- Parasitic load / MW_{th} = **0.000208**

Bad news:

- All panels lying face up on surface (no stacking) → **~53200 m^2**
- Very wind dependent



Inorganic-Organic Janus Emitter with R_{sol} ~ 90%; ε_{broad} ~ 0.85 (Se-Yeon Heo et al.)

⟹ *Very* interesting (& promising) emerging HR method!

Discussion

Validation and Verification

- Comparison between the parasitic load per MW_{th} rejected from our system against the **industry standard** was done and recorded as shown in this Table X. :

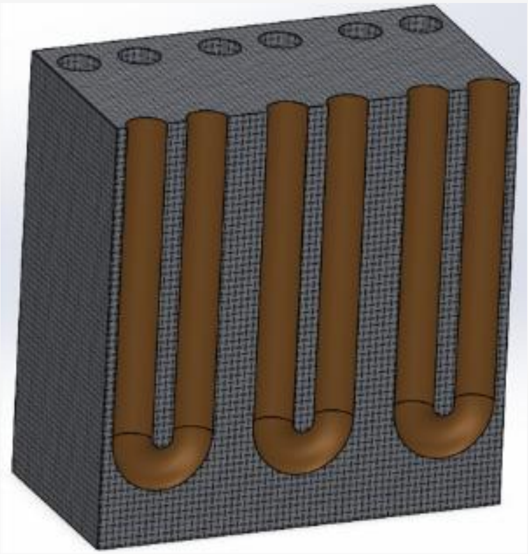
Table X. Parasitic load output compared to industry averages. (Kuzle, Bošnjak and Pandžić, n.d.)

% of Parasitic load per MW_{th} rejected		% difference
Industry standard	Imperial 5 Model	57%
4-6%	2.15%	

- Falls within the **same order of magnitude** with the industry standard value

Results are valid and verified!!

Conclusion



- There is a need for **intermediary loops** due to the radioactive He, for safety
- We want to optimise these intermediary heat exchanger loops for the criteria:

Parasitic Load

- **Thermosyphons** in porous cube
- Pump power in intermediary loop is reduced:
  - Natural convection
  - Density differences between steam and water

Footprint

- **Porous cube** as heat exchanger
- Footprint of heat exchanger is reduced:
  - Porous cube has much higher heat transfer coefficient
  - Hence, the area required is much lower

- Parasitic load of 21.5 kW per MW_{th} rejected aligns with industry benchmarks
- Reduced loop **sizing** makes the concept promising for further development

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IMPERIAL

Thank you for your attention

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